



Review of Vector and Matrix Operation (linear Algebra)

Estimated time: 20min (more, if you have forgotten how to do this and need to lookup the calculations)

As we will be using the following in the upcoming lecture, make sure you know how to perform the following calculations.

Given 2-dimensional vectors $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and 2x2-matrixes $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & 0 \\ 0 & f \end{bmatrix}$.

a) Compute $u v^T$ and $u^T v$.

$$u v^T = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 \\ u_2 v_1 & u_2 v_2 \end{bmatrix} \text{ and } u^T v \text{ is the dot-product: } u^T v = u_1 v_1 + u_2 v_2$$

b) Compute Au .

$$Au = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} au_1 + bu_2 \\ cu_1 + du_2 \end{bmatrix}.$$

c) Compute the determinantes $\det A$ and $\det B$.

$$\det A = ad - bc \text{ and } \det B = ef$$

d) Compute the inverse matrix A^{-1} and B^{-1} .

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ and therefore } B^{-1} = \begin{bmatrix} 1/e & 0 \\ 0 & 1/f \end{bmatrix}$$